

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Data set A consists of the heights of **75** objects and has a mean of **25** meters. Data set B consists of the heights of **50** objects and has a mean of **65** meters. Data set C consists of the heights of the **125** objects from data sets A and B. What is the mean, in meters, of data set C?

Correct Answer: 41

Rationale

The correct answer is **41**. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of **75** objects and has a mean of **25** meters. This can be represented by the equation $\frac{x}{75} = 25$, where x represents the sum of the heights of the objects, in meters, in data set A. Multiplying both sides of this equation by **75** yields $x = 75(25)$, or $x = 1,875$ meters. Therefore, the sum of the heights of the objects in data set A is **1,875** meters. It's also given that data set B consists of the heights of **50** objects and has a mean of **65** meters. This can be represented by the equation $\frac{y}{50} = 65$, where y represents the sum of the heights of the objects, in meters, in data set B. Multiplying both sides of this equation by **50** yields $y = 50(65)$, or $y = 3,250$ meters. Therefore, the sum of the heights of the objects in data set B is **3,250** meters. Since it's given that data set C consists of the heights of the **125** objects from data sets A and B, it follows that the mean of data set C is the sum of the heights of the objects, in meters, in data sets A and B divided by the number of objects represented in data sets A and B, or $\frac{1,875+3,250}{125}$, which is equivalent to **41** meters. Therefore, the mean, in meters, of data set C is **41**.